360 Gbps PAM4 Differentially Driven EML with 100 GHz 3-dB Bandwidth Dual Series-Connected EAMs for Next-Generation 3.2 Tbps Data Center Transceivers

Shinya Okuda[1], Yohei Hokama[1], Kenichi Terao[1], Shunto Katsumi[1], Takaya Morikawa[1], Toshiya Tsuji[1], Chikara Watatani[1], Yoshimichi Morita[1], Masaaki Shimada[1], Takuma Fujita[1], Asami Uchiyama[2], Mizuki Shirao[2], Kyosuke Kuramoto[1], Eitaro Ishimura[1], Akihito Ono[1]

[1]High Frequency & Optical Device Works, Mitsubishi Electric Corporation, 4-1, Mizuhara, Itami, Hyogo 664-8641, Japan
[2]Information Technology R & D Center, Mitsubishi Electric Corporation, 5-1-1, Ofuna, Kamakura, Kanagawa 247-8501, Japan
Author e-mail address: okuda.shinya@da.mitsubishielectric.co.jp

Abstract: Developed a high-speed differentially driven EML with dual EAMs, achieving >100 GHz 3-dB bandwidth. Demonstrated 360 Gbps (180 Gbaud PAM4) operation with a 3.3 dB TDECQ (BtB) for next-generation 3.2 Tbps data center transceivers. © 2026 The Author(s)

1. Introduction

The widespread adoption of video streaming services and rapid advancements in artificial intelligence (AI) and machine learning (ML) have led to significant growth in data traffic within data centers in recent years. Consequently, there is a growing demand for higher transmission speeds in data center transceivers. Currently, 800 Gbps and 1.6 Tbps transceivers, which integrate multiple 200 Gbps optical devices, are being actively developed [1]. However, next-generation 3.2 Tbps transceivers will require optical devices operating beyond 200 Gbps. Various modulators capable of high-speed modulation, such as silicon photonics modulators and thin-film lithium niobate modulators, are under active research [2,3]. Among these, electro-absorption modulated lasers (EMLs) have become a dominant technology for intensity-modulation direct-detection (IM-DD) transmitters in data centers due to their wide bandwidth, low power consumption, and compact footprint. Several studies have reported achieving over 200 Gbps using EMLs operating beyond 100 Gbaud with N-level pulse amplitude modulation (PAM4, PAM6, PAM8) [4-9], thereby establishing EMLs as promising candidates for next-generation 3.2 Tbps transceivers.

Our previous work [9] focused on high-speed EMLs with single-ended driving. However, as modulation speeds increase, electrical crosstalk between densely integrated lanes becomes a critical issue. To enhance noise immunity, differential driving of EMLs is increasingly required [10,11]. In this work, we report the development of a high-speed differentially driven EML with a dual electro-absorption modulator (EAM) structure. We achieved a 3-dB bandwidth exceeding 100 GHz and demonstrated transmitter and dispersion eye closure quaternary (TDECQ) values below 3.4 dB for 360 Gbps (180 Gbaud PAM4) in a back-to-back (BtB) configuration, as well as for 340 Gbps (170 Gbaud PAM4) after 500 m transmission. These results are anticipated to significantly advance the realization of next-generation 3.2 Tbps transceivers.

2. Design of the EML

Figure 1 illustrates the schematic structure of the developed differentially driven EML chip. This EML features a unique hybrid waveguide structure, monolithically integrating a buried heterostructure (BH) distributed feedback laser diode (DFB-LD), a high-mesa EAM waveguide, and a spot-size converter (SSC) on a single substrate. A BH structure is utilized in the DFB-LD to achieve high optical output power. The high-mesa EAM waveguide enables lower capacitance and higher optical confinement due to its larger horizontal refractive index step compared to buried or ridge-type waveguides [12]. The SSC enhances the fiber coupling efficiency.

In conventional single-ended driven EMLs, common cathode electrodes are typically used for the LD and EAM. In contrast, this EML features an LD and EAMs with electrically isolated cathode electrodes to enable differential driving. Differential driving typically reduces the bandwidth by half compared to single-ended driving, due to the 100-Ω termination resistance. Our device adopts a dual-EAM structure. By connecting two EAMs in series, the capacitance can be halved compared to a single EAM of the same length, which enables high-speed operation even with differential driving. Furthermore, connecting the two EAMs to a common ground (GND) electrode enables the application of independent biases to each EAM.

For characterization, the EML chip was mounted in a chip-on-carrier (CoC) configuration. The EML chip was mounted using conventional junction-up bonding and bonding wires. The termination resistance was 100 Ω (50 Ω × 2).

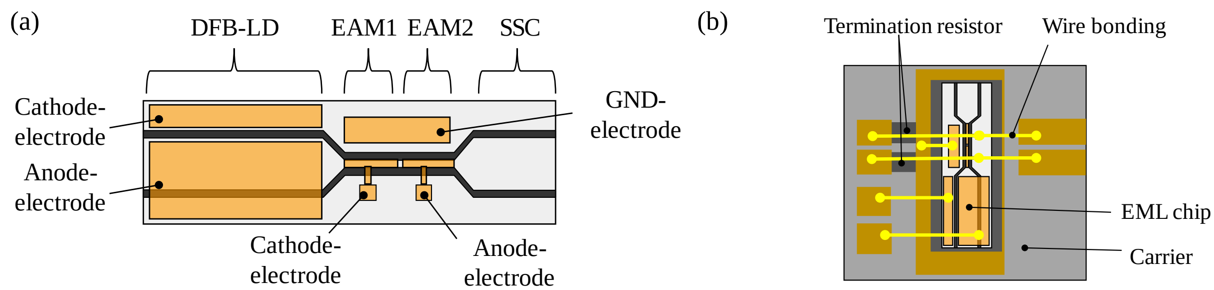

Fig. 1. Schematic structure of (a) the EML chip and (b) the CoC configuration.

3. Experimental result

All measurements were performed in a CoC configuration at 55°C, with the EML temperature precisely controlled by a thermoelectric cooler (TEC). The lasing wavelength was 1.31 µm. Figure 2(a) shows the light-current (LI) characteristics. The threshold current was 8 mA, and an output power of 19 mW was obtained at 100 mA. Figure 2(b) shows the electro-optic (EO) frequency response (S21) measurement results. The electrical signal from the vector network analyzer (VNA, Anritsu ME7838AX) was applied to the EML via a bias-tee, coaxial cables, and an RF probe. The modulated optical output from the EML was converted to an electrical signal by an optical-to-electrical converter (Anritsu MN4765B) connected to the VNA. Subsequently, the differential S21 characteristics were calculated from the S-parameters of the two EAMs. Regarding biasing, +2.3 V and -2.3 V were applied to the two EAMs, respectively, via a bias-tee, while a DC operating current of 100 mA was supplied to the DFB-LD. A 3-dB bandwidth exceeding 100 GHz was achieved, primarily due to the low-capacitance dual EAM structure.

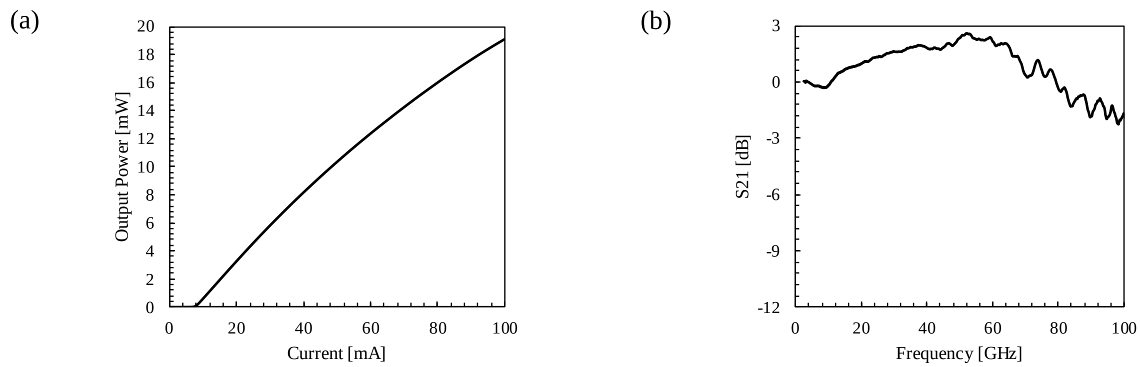

Fig. 2. (a) Light-current (LI) characteristics and
(b) Electro-optic (EO) frequency response (S21) at 55°C with a DFB-LD current of 100 mA.

PAM4 transmission characteristics were evaluated over a range of 100 Gbaud to 180 Gbaud. An arbitrary waveform generator (AWG, Keysight M8199B) generated a differential electrical PAM4 short stress pattern random quaternary (SSPRQ) signal, which was pulse-shaped using a raised-cosine filter. This signal was then amplified by an electrical amplifier, with its peak-to-peak differential modulation amplitude (Vppd) set to 1.5 V. After transmission over various fiber lengths (BtB, 500 m, and 2000 m), the optical output from the EML, operating without optical amplification, was coupled to a sampling oscilloscope (Keysight N1032A) equipped with a 120 GHz 4th-order Bessel-Thomson filter. The received signal was equalized using a 15-tap T-spaced feed-forward equalizer (FFE), and the TDECQ was subsequently measured. The TDECQ values were calculated at a target symbol error ratio (SER) of 9.6×10^{-3} corresponding to the KP4 forward error correction (FEC) code with soft-decision concatenated FEC, as specified in IEEE P802.3dj [1]. The EML driving conditions were identical to those used for the frequency response measurements. Figure 3 (a) presents the optical waveforms for BtB and 500 m transmission at 340 Gbps (170 Gbaud) and 360 Gbps (180 Gbaud). Clear eye diagrams were observed. The extinction ratio (ER) was approximately 4 dB, satisfying the ER specification defined in IEEE P802.3dj. Figure 3 (b) illustrates the baud rate dependence of TDECQ. In the BtB configuration, TDECQ values below 3.4 dB were achieved up to 360 Gbps

Optical Fiber Communication Conference (OFC) © 2026 Optica Publishing Group

(180 Gbaud). Furthermore, the same TDECQ criterion was met after 500 m transmission up to 340 Gbps (170 Gbaud) and after 2000 m transmission up to 320 Gbps (160 Gbaud).

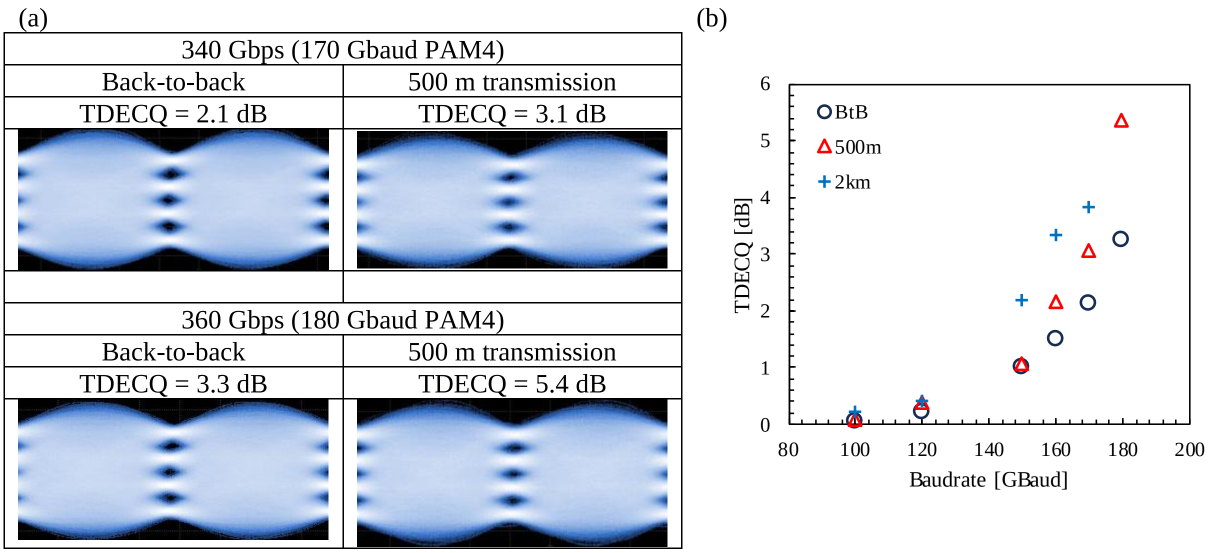

Fig. 3. (a) Optical waveforms of the EML and (b) baudrate dependence of PAM4 TDECQ values after equalization with a 15-tap T-spaced FFE.

4. Conclusion

We developed a high-speed differentially driven EML and experimentally demonstrated operation exceeding 200 Gbps. To achieve such high-speed performance even with differential driving, a dual EAM structure comprising two series-connected EAMs was adopted, which resulted in a 3-dB bandwidth exceeding 100 GHz. A TDECQ of 3.3 dB was achieved at 360 Gbps (180 Gbaud PAM4) in a BtB configuration. Furthermore, TDECQ below 3.4 dB was achieved after 500 m transmission up to 340 Gbps (170 Gbaud), and after 2000 m transmission up to 320 Gbps (160 Gbaud). We anticipate that operation exceeding 400 Gbps could be achievable through further improvements in electrical waveform quality and optimized implementation schemes. These results are expected to significantly contribute to the development of next-generation 3.2 Tbps transceivers for data centers.

5. References

[1] IEEE P802.3dj 200 Gb/s, 400 Gb/s, 800 Gb/s, and 1.6 Tb/s Ethernet Task Force, https://www.ieee802.org/3/dj/index.html (2024)
[2] A. Ostrovskis, et al., "Traveling-Wave Silicon Photonics Mach-Zehnder Modulator for Beyond 350 Gb/s Transmission in C-band," M1G.2, OFC 2025
[3] A. Cui, et al., "Integrated Optical Transceivers for Data Center Applications," M1G.3, OFC 2025
[4] H. Asakura et al., "420 Gbps PAM8 Operation Using 93 GHz Bandwidth Lumped-Electrode Type EA-DFB Laser at 50°C beyond 400 Gbps/lane," Th1E.3, ECOC 2022.
[5] O. Ozolins et al., "Optical Amplification-Free 310/256 Gbaud OOK, 197/145 Gbaud PAM4, and 160/116 Gbaud PAM6 EML/DML-based Data Center Links," Th4B.2, OFC 2023
[6] A. Ostrovskis et al., "Heterogenous InP Electro-Absorption Modulator with Si Waveguides for Beyond 200 Gbps/λ Optical Interconnects," Th4C.8, OFC 2024
[7] A. Uchiyama et al., "Demonstration of 155 Gbaud PAM4 and PAM6 EML with Narrow High-Mesa EA Modulator for 400 Gbps per Lane Transmission," Tu2D.1, OFC 2024
[8] H. Asakura et al., "400 Gbps Semi-cooled Lumped-Electrode EA-DFB Laser Operating at 200 Gbaud PAM4 Modulation with 0.6 Vpp," Th3A.1, ECOC 2024
[9] S. Okuda, et al., "High-Speed 340 Gbps PAM4 and 450 Gbps PAM6 Operations of Narrow High-Mesa EML," Tu2J.7, OFC 2025
[10] Z. Zhang et al., "Differential Drive EML with P/N Electrode Isolated toward Low Vpp Applications," M3K.6, OFC 2025
[11] S. Ohno et al., "226 Gbps PAM4 Operation Using Differential Drive EA-DFB Laser with 2.0-Vppd Swing over 10-km SSMF Transmission for 1.6TbE Transceivers," M4K.6, OFC 2025
[12] Y. Morita et al., "1.3 µm 28 Gb/s EMLs with Hybrid Waveguide Structure for Low-Power-Consumption CFP2 Transceivers," OFC 2013